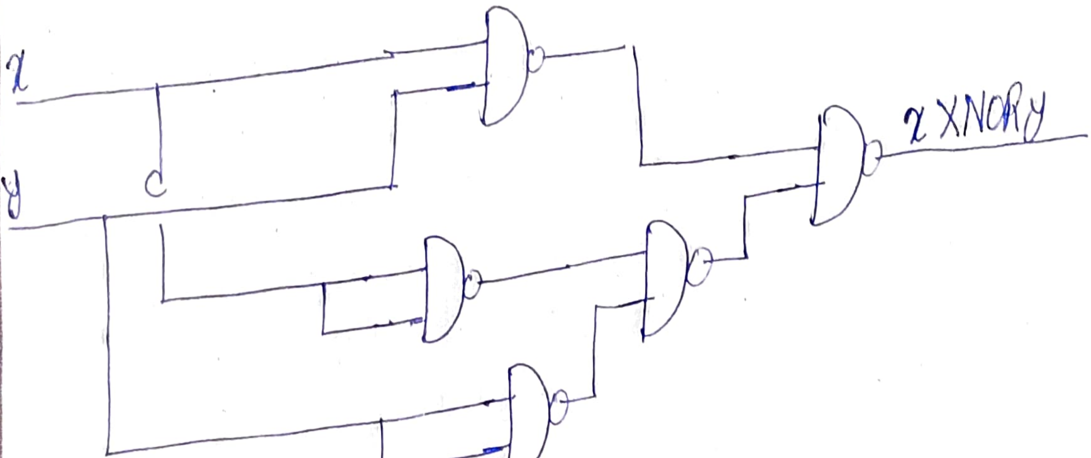
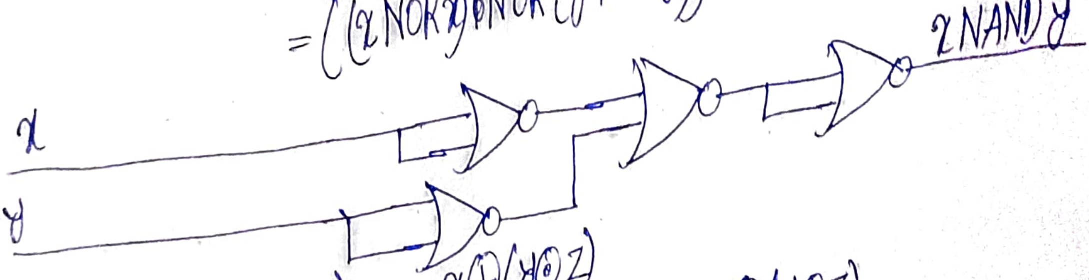
2-25: Implement Ex-NOR gate using minimum number of NAND gate and NAND gates using minimum number of NOR gates.

Solution: The Boolean expression of XNOR gate is $x \cdot y + x' \cdot y'$.

$$x \cdot y + x' \cdot y' = ((x \cdot y + x' \cdot y')')' \quad \text{(Theorem 3)}$$
$$= ((x \cdot y)' \cdot (x' \cdot y')')' \quad \text{(Theorem 5(a))}$$
$$= ((x \cdot y)' \cdot ((x \cdot x)' \cdot (y \cdot y)')')' \quad \text{(Theorem 1(b))}$$
$$= ((x \text{ NAND } y) \text{ NAND } ((x \text{ NAND } x) \text{ NAND } (y \text{ NAND } y)))$$



$$x \text{ NAND } y = (x \cdot y)' = x' + y' \quad \text{(Theorem 5(b))}$$
$$= ((x' + y')')' \quad \text{(Theorem 3)}$$
$$= ((x + x)' + (y + y)')')' \quad \text{(Theorem 1(a))}$$
$$= ((x \text{ NOR } x) \text{ NOR } (y \text{ NOR } y))'$$



2-26: Show that $(x \oplus y) \odot z = x \oplus (y \odot z)$

x	y	z	$x \oplus y$	$(x \oplus y) \odot z$	$y \odot z$	$x \oplus (y \odot z)$
0	0	0	0	1	1	1
0	0	1	0	0	0	0
0	1	0	1	0	0	0
0	1	1	1	1	1	1
1	0	0	1	0	1	0
1	0	1	1	1	0	1
1	1	0	0	1	0	1
1	1	1	0	0	1	0